Juliana Gomes Ferreira de Souza

Curriculum Vitae

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Research Profile

Applied mathematician working in coding theory, information theory, and lattice-based methods for communications. My research focuses on algebraic constructions of lattice codes, multilevel coding schemes, decoding algorithms, and rigorous performance analysis. I combine algebraic methods, lattice geometry, and reproducible computational experiments, with interests in synchronisation-error channels, federated learning, and wireless communication systems.

Research Experience

Postdoctoral Fellow *December 2025–present*
Brazilian Institute of Data Science (BI0S), University of Campinas (UNICAMP), Brazil
Institute of Mathematics, Statistics and Scientific Computing (IMECC)
Research: federated learning, information theory, and lattice coding.
Supervisor: Sueli I. R. Costa. Funding: FAPESP.

Research Visit *January–March 2023*
Imperial College London, Department of Electrical and Electronic Engineering, UK
Research: lattice codes, quaternion algebras, and coding and decoding methods.
Supervisor: Cong Ling. Funding: Santander/DERI.

Education

PhD in Applied Mathematics *2019–2025*
University of Campinas (UNICAMP), Brazil
Thesis: *Lattice Coding for Reliable and Efficient Communication.*
Supervisor: Sueli I. R. Costa. Scholarship: CAPES.

Certificate in Distance Education Teaching *2023–2025*
Virtual University of the State of São Paulo (UNIVESP), Brazil
Final paper: *Challenges and Potentialities of Project-Based Learning in Distance Education at UNIVESP*. Supervisor: José Ricardo Tarpani.

MSc in Applied and Computational Mathematics *2017–2019*
University of Campinas (UNICAMP), Brazil
Dissertation: *The Sphere Packing Problem in n-Dimensional Space.*
Supervisor: Sueli I. R. Costa.

BSc in Mathematics *2006–2010*
Federal University of Pará (UFPA), Brazil
Undergraduate thesis: *The Geometry of DNA.*
Supervisor: Marcos Monteiro Diniz.

Publications

Journal Articles

- J. G. F. Souza, S. I. R. Costa, and C. Ling. *Multilevel Lattice Codes From Hurwitz Quaternion Integers*. IEEE Transactions on Information Theory, 71(11), 8280–8293, November 2025. DOI: 10.1109/TIT.2025.3603487. arXiv:2401.10773.
- J. G. F. Souza and S. I. R. Costa. *Achieving Uniform Side Information Gain With Multilevel Lattice Codes Over the Ring of Integers*. IEEE Communications Letters, 29(7), 1739–1743, July 2025. DOI: 10.1109/LCOMM.2025.3573236. arXiv:2501.14921.

Conference Proceedings and Abstracts

- J. G. F. Souza and Y.-C. Huang. *Finite-Blocklength Analysis of Alamouti Codes over Eisenstein Integers*. 2026 IEEE International Symposium on Information Theory (ISIT), Guangzhou, China, pp. 1–6, 2026. DOI: 10.1109/ISIT62367.2026.11653867. arXiv:2604.10137.
- J. G. F. Souza and S. I. R. Costa. *Lattice Index Coding from Construction π_A Lattices*. Proceedings of CNMAC 2024, Porto de Galinhas, Brazil. Proceeding Series of the Brazilian Society of Computational and Applied Mathematics, 11(1), 2025. DOI: 10.5540/03.2025.011.01.0420.
- J. G. F. Souza, S. I. R. Costa, and C. Ling. *Construction π_A Lattices Extended to Hurwitz Quaternion Integers*. 2024 IEEE International Symposium on Information Theory (ISIT), Athens, Greece, pp. 2186–2191, 2024. DOI: 10.1109/ISIT57864.2024.10619280.
- J. G. F. Souza and S. I. R. Costa. *Decodificação multicamada em códigos reticulados* [Multilayer Decoding in Lattice Codes]. CNMAC 2023, Bonito, Brazil. Proceeding Series of the Brazilian Society of Computational and Applied Mathematics, 10(1), 2023. Conference abstract. Proceedings.
- J. G. F. Souza and S. I. R. Costa. *Sobre Anéis de Inteiros e Aplicações em Comunicações* [On Rings of Integers and Applications in Communications]. Proceedings of the X Bienal da Matemática, Brazilian Mathematical Society (SBM), p. 318, 2022. Proceedings.

Preprints

- J. G. F. Souza and Y.-C. Huang. *Shaping and Finite-Blocklength Analysis of Quaternionic Alamouti-Type Codes*. arXiv:2610.41693, 30 October 2026.
- J. G. F. Souza, C. Li, and C. Ling. *Construction π_A over Multiquadratic Fields for Compound Block-Fading Wiretap Channels*. arXiv:2604.12703, 2026. Accepted for presentation at WCC 2026.

Teaching Experience

Teaching Assistant, University of Campinas *2019–2024*
Graduate level: Professional Master's Programme in Applied and Computational Mathematics

- *Vector Structures* (2019–2024). Assisted Sueli I. R. Costa in remote and in-person courses across multiple semesters. Responsibilities included tutoring students, creating assignments, and guiding project work.
- *Geometric Analysis of Multivariable Functions* (2019–2023). Assisted Sueli I. R. Costa, Priscila C. B. Rampazzo, and João E. Strapasson. Supported problem-solving sessions, evaluated reports, and provided feedback.

Undergraduate level: Interdisciplinary Higher Education Programme

- *Basic Mathematics I and II* (2022). Assisted Francisco A. M. Gomes Neto through the Teaching Internship Programme (PED). Liaised between the professor and students, conducted review sessions, and supported students' transition to university-level mathematics.
- *Linear Algebra* (2021). Assisted Sueli I. R. Costa through PED, including reviewing proofs, mentoring students, and facilitating group problem-solving sessions.

Academic Tutor, UNIVESP *2023–2025*
Virtual University of the State of São Paulo, Brazil

- Supported asynchronous and synchronous learning in STEM programmes, including *Object-Oriented Programming*, *Data Structures*, *Databases*, and *Applications in Machine Learning*.
- Guided student projects in computing (2024) and education (2025), provided individual feedback, and helped students manage their learning in a fully online environment.

Awards

- **Marco Antônio Raupp Award**, 2026. Best PhD Thesis, Brazilian Society of Computational and Applied Mathematics (SBMAC).
- **Best PhD Thesis in Applied Mathematics**, 2026. IMECC, University of Campinas, for a thesis defended in 2025.

Academic Service

Member of the organising team for *EnCoRI 2023 – Encontro de Códigos, Reticulados e Informação* (Meeting of Codes, Lattices and Information), UNICAMP.

Technical Skills and Languages

Scientific computing: Wolfram Mathematica, SageMath, Python.
Languages: Portuguese (native); English (C1, advanced).

Conferences and Scientific Meetings

2026

- SECRET – Internal Seminar of the Lattices and Codes Research Group. IMECC, UNICAMP, Campinas, Brazil, 26–27 October.
- III BI0S Workshop. Galileu Laboratory, UNICAMP, Campinas, Brazil, 22–23 October.
- National Congress of Applied and Computational Mathematics (CNMAC). Presidente Prudente, Brazil, 14–18 September.
- IEEE International Symposium on Information Theory (ISIT). Guangzhou, China, 28 June–3 July.
- Workshop on Coding and Cryptography (WCC). Paris, France, 8–12 June.
- Inaugural Workshop of the Brazilian Center for Geometry (CBG). Campinas, Brazil, 17–20 March.
- CIMPA School on Algebraic Geometric Methods in Information Theory (AGMIIT). Santa Fe, Argentina, 2–13 March.
- II Week of Arithmetic Meetings (SemEAr). Campinas, Brazil, 24–27 February.

Earlier Meetings

- 16th International Conference on Finite Fields and Their Applications (Fq16). São Carlos, Brazil, 2025.
- International Symposium on Information Theory and Its Applications (ISITA). Taipei, Taiwan, 2024.
- National Congress of Applied and Computational Mathematics (CNMAC). Porto de Galinhas, Brazil, 2024.
- XXVII Brazilian Algebra Meeting. São Paulo, Brazil, 2024.
- IEEE International Symposium on Information Theory (ISIT). Athens, Greece, 2024.
- National Congress of Applied and Computational Mathematics (CNMAC). Bonito, Brazil, 2023.
- Meeting of Codes, Lattices and Information (EnCoRI). Campinas, Brazil, 2023.
- X Bienal da Matemática. Belém, Brazil, 2022.
- Pan-Amazonian Mathematics Congress. Belém, Brazil, 2018.
- International Congress of Mathematicians (ICM). Rio de Janeiro, Brazil, 2018.
- I Fractal Geometry Exposition. Belém, Brazil, 2009.